

Visual Product Concept: Carbon Footprint Awareness Platform

"Making Green Habits Visual, Relatable, and Fun"

This document translates our technical system architecture into a visual roadmap, user journey, and simple screens. Use this to present the product vision to designers, investors, marketing teams, and non-technical stakeholders.

1. The Core Vision (The 30-Second Pitch)

Most environmental tools fail because they are too clinical, asking for complex utility bill numbers that users don't have.

Our platform solves this by making sustainability **frictionless, relatable, and social**. Instead of focusing purely on dry carbon math ($kgCO_2e$), we convert raw data into friendly daily checklists, visual charts, and tangible real-world comparisons (like "Smartphones Charged" or "Trees Planted").

[FAST ONBOARDING] ---> [VISUAL DASHBOARD] ---> [ATOMIC ACTIONS] ---> [SOCIAL REWARDS]

(60-Sec Baseline Quiz) (Translates CO2 into (Bite-sized daily (Turn points into
relatable visuals) habit checklists) real planted trees)

2. The User Journey: From "Eco-Curious" to "Eco-Champion"

To make the app sticky, we guide the user through a clear, rewarding progression path:

Stage 1: Discover	Stage 2: Understand	Stage 3: Transform
+-----+ +-----+ +-----+		
"How big is my footprint?"	"What is causing it?"	"What can I do today?"
• Quick 60-sec onboarding	• Visual pie charts of	• Daily 1-tap checklists
• Localized starting state	lifestyle categories	• Social Group Challenges
• Instant personal goal	• Easy-to-understand	• Redeem points for real
setting (e.g. -20%)	eco-comparisons	reforestation projects
+-----+ +-----+ +-----+		

3. Visual Storyboard: Key App Screens

Below are visual wireframe mockups showcasing how the primary screens should look and feel to an end-user.

Screen A: Frictionless Onboarding (The Setup)

Designed to bypass questionnaire fatigue with visual sliders.

```
+-----+
|               STEP 2 of 4               |
|                                         |
|      How do you usually get around?      |
|                                         |
|  [🚗 Car]  [🚌 Transit]  [🚲 Bike/Walk]  |
|                                         |
|  <---[=====●=====]>--->|
|  0 km/week                      150+ km/week |
|                                         |
|      ( 80 km )                      |
|                                         |
|  +-----+ |
|  | 💡 Did you know? Traveling 80 km by car | |
|  | creates about $14.4 \text{ kg of } CO_2$. | |
|  +-----+ |
|                                         |
|  [ Back ]                      [ Next ] |
+-----+
```

Screen B: The Daily Dashboard (The Hub)

Replaces dry graphs with a central, highly relatable carbon progress meter.

```
+-----+
| [👤 Profile]   ECO-DASHBOARD   [🔔 3] |
+-----+
| "Excellent work, Alex! You are beating your |
| weekly target by 12%."                      |
|                                         |
|      /=====\                          |
|      / 18.5 kg CO_2e \                   |
|      ( Saved This )                      |
|      \ Week!   /                        |
+-----+
```

```

|           \=====|
|
| That is equivalent to:
| 🌳 1 mature tree absorbing carbon for a year
| 📱 2,250 smartphone battery charges
|
| MY PROGRESS THIS WEEK
| [=====●-----] Target
| (Spent: 42 kg / Budget: 65 kg)
|
| [➕ Quick-Log Activity] [🏆 Join Weekly Challenge]
+-----+
| [🏠 Home] [📊 Insights] [✅ Habits] [🎁 Shop] |
+-----+

```

Screen C: Daily Habit Checklist (The Action Engine)

Where real behavioral change happens. Keeping it low-effort and positive.

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+-----+
| [<- Back]      DAILY HABITS      [🔥 5] |
+-----+
| Complete today's habits to protect your 5-day
| streak and earn eco-points!
|
| TODAY'S CHECKLIST
| [X] Commuted without a car          +15 pts |
|   "Awesome! You saved $3.2 \text{ kg of } CO_2$."|
|
| [ ] Eat a plant-based meal          +10 pts |
|   (Beef alternatives save over $5 \text{ kg}$!) |
|
| [ ] Wash clothes on "cold cycle"    +5 pts |
|
| [ ] Avoided single-use plastics      +5 pts |
|
| -----|
| WEEKLY LEADERBOARD (Eco-Friends)
| 1. Sarah M. 🥇 120 pts (Saved $24 \text{ kg}$) |
| 2. You 🥈 105 pts (Saved $18 \text{ kg}$) |
| 3. Daniel K. 🥉 90 pts (Saved $15 \text{ kg}$) |
+-----+

```

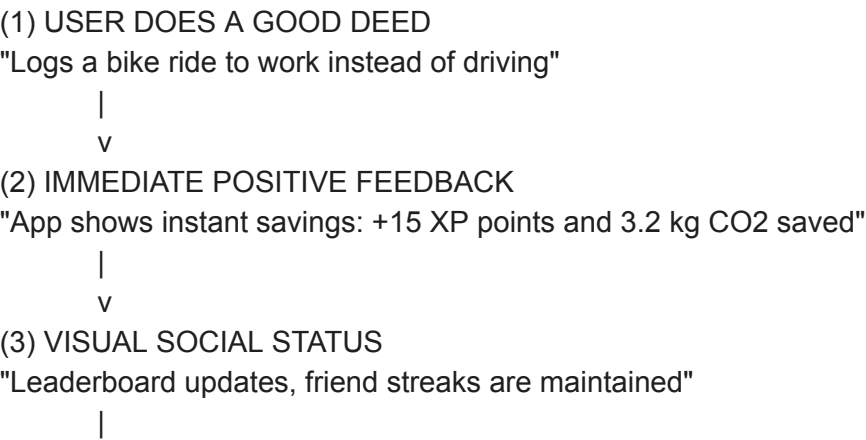
4. The Eco-Equivalents Engine (Making Math Human)

A non-technical user cannot conceptualize what "1 kg of CO₂e" looks or feels like. The app relies heavily on a translation system to instantly contextualize carbon data:

Raw Data Value	Human-Friendly Visual Translation
1 kg of CO ₂	Charging a smartphone 122 times.
5 kg of CO ₂	Driving a gas-powered car for 21 km.
22 kg of CO ₂	The total workload of 1 mature tree over an entire year.
150 kg of CO ₂	Keeping an average house lightbulb lit continuously for 1 year.
1,000 kg of CO ₂ (1 ton)	A one-way flight from New York to London for one passenger.

5. The "Good Habits" Reward Loop (How We Retain Users)

Instead of guilt-tripping users about climate change, our design centers entirely on a **positive reinforcement loop**:



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(4) TANGIBLE IMPACT REWARDS

"User redeems points in the Shop to plant a real tree in Kenya"

6. Real-World Integration Story (The "Magic" Features)

To help stakeholders understand how automated tracking fits seamlessly into the user's actual life:

- **The Morning Commute:** The app syncs quietly with fitness APIs (like Apple Health or Google Fit). If it detects the user walked or biked 3 km instead of driving, it automatically pops up a notification: *"We detected active travel! You just saved $0.8\text{ kg of }CO_2$."*
- **The Grocery Trip:** In later phases, a simple photo scan of a grocery receipt automatically extracts the protein choices (e.g., Tofu vs. Beef) and populates the diet log in 2 seconds.
- **The Electric Bill:** Users can securely link their smart home energy accounts once. From then on, the app automatically tracks grid-level electricity drops and rewards them when they use energy during off-peak, green hours.